Downloaded from geology.gsapubs.org on October 28, 2010

Geology

Magnetostratigraphy of displaced Upper Cretaceous strata in southern California

J. Gilbert Fry, David J. Bottjer and Steve P. Lund

Geology 1985;13;648-651
doi: 10.1130/0091-7613(1985)13<648:MODUCS>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geology

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from geology.gsapubs.org on October 28, 2010

Magnetostratigraphy of displaced Upper Cretaceous strata in southern California

J. Gilbert Fry, *David J. Bottjer, Steve P. Lund
Department of Geological Sciences, University of Southern California, Los Angeles, California 90089-0741

ABSTRACT

A magnetostratigraphic study of Upper Cretaceous marine strata from the Santa Ana Mountains in southern California has identified a Campanian reversed magnetozone. This reversed interval, corresponding to marine magnetic anomaly 33-34 (Chron 33r) of Campanian age, can be correlated with a Campanian reversed magnetozone that has been reported from strata of the Great Valley Sequence in central California. The Late Cretaceous paleolatitude of the Santa Ana Mountains is estimated from this study to be 26.6°N. This is significantly different from the region's expected Cretaceous paleolatitude of 43.8°N as part of the North American stable craton, and indicates that this region (part of the Peninsular Ranges terrane) was 1900 km farther south in Cretaceous time relative to the stable craton.

INTRODUCTION

Exact dating and correlation of Cretaceous stratigraphic sequences in California is difficult because of the discontinuous nature of outcrops, general similarity of lithofacies, and common scarcity of biostratigraphically diagnostic fossils. The development of a magnetic polarity time scale for the Late Cretaceous (e.g., Lowrie and Alvarez, 1981; Harland et al., 1982) has provided an alternative methodology to date and correlate these rocks. Ward et al. (1983) identified the lower Campanian reversed magnetozone in four sections of marine strata from the California Great Valley Sequence (Fig. 1a), and have demonstrated the usefulness of this approach. The primary purpose of this study has been to initiate development of a detailed magnetic polarity stratigraphy for Upper Cretaceous strata in southern California. We have concentrated our initial studies on the classic stratigraphic sections exposed in the Santa Ana Mountains (Fig. 1a), which previous studies have shown to be both more fossiliferous and more stratigraphically complete than any other exposed Upper Cretaceous sections in this region (Popenoe, 1942; Bottjer and Link, 1984).

At the same time, our interest in paleogeographic reconstructions of Upper Cretaceous strata from California (Bottjer and Link, 1984) has led us to study the possible allochthonous nature of some of these strata. The concept that much of western North America is composed of allochthonous tectonostratigraphic terranes has already played an important role in several recent paleogeographic and tectonic reconstructions (e.g., Jones et al., 1983; Vedder et al., 1983). Paleomagnetic results obtained from the Upper Cretaceous strata in the Santa Ana Mountains should provide important new information concerning the possible displacement of terranes in which these rocks are located.

GEOLOGIC SETTING

The Santa Ana Mountains are located in the northern part of the Peninsular Ranges (Peninsular Ranges terrane of Bottjer and Link, 1984) and border the southeastern margin of the Los Angeles Basin (Fig. 1a). The Upper Cretaceous rocks in the Santa Ana Mountains comprise a thick (500- to 1500-m) succession of predominantly marine terrigenous clastics that lie unconformably between strata of Triassic?–Early Cretaceous and Paleocene ages (Fig. 1b) (Schoellhamer et al., 1981; Bottjer and Link, 1984). The Upper Cretaceous strata crop out as a northwest-striking homocline that dips west to southwest at 15° to 40° (Schoellhamer et al., 1981).

Sampling for this study was conducted at Silverado Canyon in the northern Santa Ana Mountains (Fig. 1a), where Upper Cretaceous rocks of the Ladd and Williams Formations are exposed in road and stream cuts. The conglomerate, sandstone, mudstone, shale, and limestone that compose these units were deposited in several different environments. These include fan delta (lower part of the Baker Canyon Member, parts of the Schulz Member), shallow marine (upper part of the Baker Canyon Member, parts of the Schulz and Pleasants Members), shelf (lower and some upper parts of the Holz Shale Member, parts of the Pleasants Member), and slope (central part of the

*Present address: Mobil Oil Corp., P.O. Box 900, Dallas, Texas 75221.

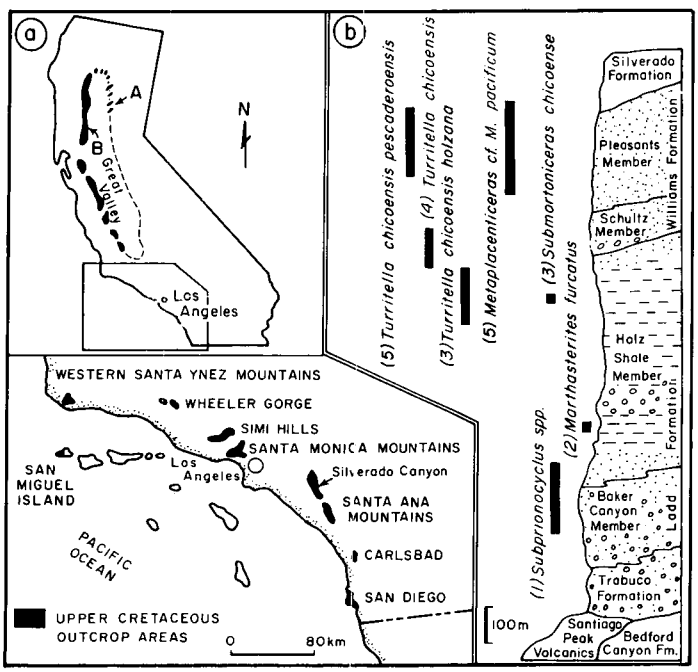


Figure 1. a: Outcrop maps for Upper Cretaceous strata in Great Valley (Great Valley Sequence) and southern California. A (Chico Creek, Butte Creek) and B (Sand Creek, Salt Creek) indicate localities where Ward et al. (1983) have identified marine magnetic anomaly 33-34. b: Idealized columnar section for Upper Cretaceous, underlying and overlying formations in northern Santa Ana Mountains, showing approximate stratigraphic position of age-diagnostic fossils. Santiago Peak Volcanics are Late Jurassic to Early Cretaceous, Bedford Canyon Formation is Triassic? to Jurassic, and Silverado Formation is Paleocene in age; unfossiliferous Trabuco Formation is thought to have a Turonian age based on gradational contact with Ladd Formation (see review in Bottjer and Link, 1984). Ages indicated by fossils are (1) late Turonian; (2) early late Coniacian to early early Campanian; (3) early Campanian; (4) middle Campanian; and (5) late Campanian. Modified from Saul (1983), Ward et al. (1983), and Bottjer and Link (1984).

Downloaded from geology.gsapubs.org on October 28, 2010

Holz Shale Member) (Almgren, 1973, 1982; Sundberg and Cooper, 1978; Saul, 1982; Buck, 1983). Deposition of the Ladd and Williams Formations was strongly influenced by the Late Cretaceous eustatic sea level rise (Bottjer and Link, 1984).

Age determinations of these strata have previously been made on the basis of occurrence of a number of geographically widespread macrofossils and microfossils. The Baker Canyon Member and the basal Holz Shale Member contain the ammonites *Subprionocyclus normalis* and *S.* cf. *S. neptuni* (Fig. 1b), indicating that Ladd Formation deposition began by at least late Turonian time (Matsumoto and Popenoe *in* Matsumoto, 1960; Popenoe et al., 1960; Saul, 1982). *S. neptuni* is known worldwide from upper Turonian strata (Matsumoto, 1959). Presence of the calcareous nannofossil *Marthasterites furcatus* in the lower part of the Holz Shale Member (Fig. 1b; Filewicz *in* Almgren, 1982) indicates an age of early late Coniacian to early early Campanian (Sissingh, 1977); *M furcatus,* although absent in the Coniacian and Campanian stratotypes in France, has an otherwise worldwide distribution (Sissingh, 1977, 1978). The ammonite *Submortoniceras chicoense,* recorded from the upper part of the Holz Shale Member (Fig. 1b), dates this part of the sequence as early Campanian (Saul, 1982, 1983). *Submortoniceras* is considered to be a worldwide indicator of early to middle Campanian age strata (Popenoe et al., 1960; Ward et al., 1983). The Pleasants Member contains the ammonite *Metaplacenticeras* cf. *M. pacificum,* which indicates that these strata are late Campanian in age (Matsumoto, 1959; Popenoe et al., 1960; Saul, 1982, 1983). Studies of the stratigraphic occurrence of fossils with more regional distributions, such as species and subspecies of the gastropod *Turritella* (Fig. 1b; Saul, 1983) and foraminifera (Almgren, 1982) support these age determinations.

PALEOMAGNETIC PROCEDURES

Block or 2.5-cm-diameter core samples were collected from each of 49 sites in the Ladd and Williams Formations (7.2-m average sampling interval) for paleomagnetic analysis. Sampled rock types included sandstone, mudstone, and shale, as well as limestone concretions of early diagenetic origin (Buck, 1983). In the laboratory, four 2-cm^3 specimen cubes were cut from the blocks, and four 2.5-cm-long cylinders were cut from the cores. The resulting paleomagnetic specimens were stepwise thermally demagnetized at 50 °C intervals up to 400–600 °C and measured in a cryogenic magnetometer.

Thermal demagnetizations indicated that a single magnetic mineral with a Curie temperature of 400–450 °C carries the natural remanent magnetization (NRM) in all the specimens. In many cases, a secondary normal-polarity (viscous?) magnetization was removed by 200 °C. Zijderveld diagrams (Figs. 2a, 2b) were used to estimate the characteristic remanence (above 200 °C) and polarity for each site. The sites were then classified in the following manner: Class 1 sites (N = 20) contained well-defined field vectors (characteristic remanences) with good polarity information, Class 2 sites (N = 25) contained a good record of polarity with less well-defined field vectors, and Class 3 sites (N = 4) contained an ambiguous record of mixed polarity and were unsuitable for further study.

MAGNETOSTRATIGRAPHY

Figure 3 shows the lithostratigraphy and corresponding magnetostratigraphy of Upper Cretaceous strata exposed in Silverado Canyon. The Ladd and Williams Formations are primarily normal in polarity, but a large reversed

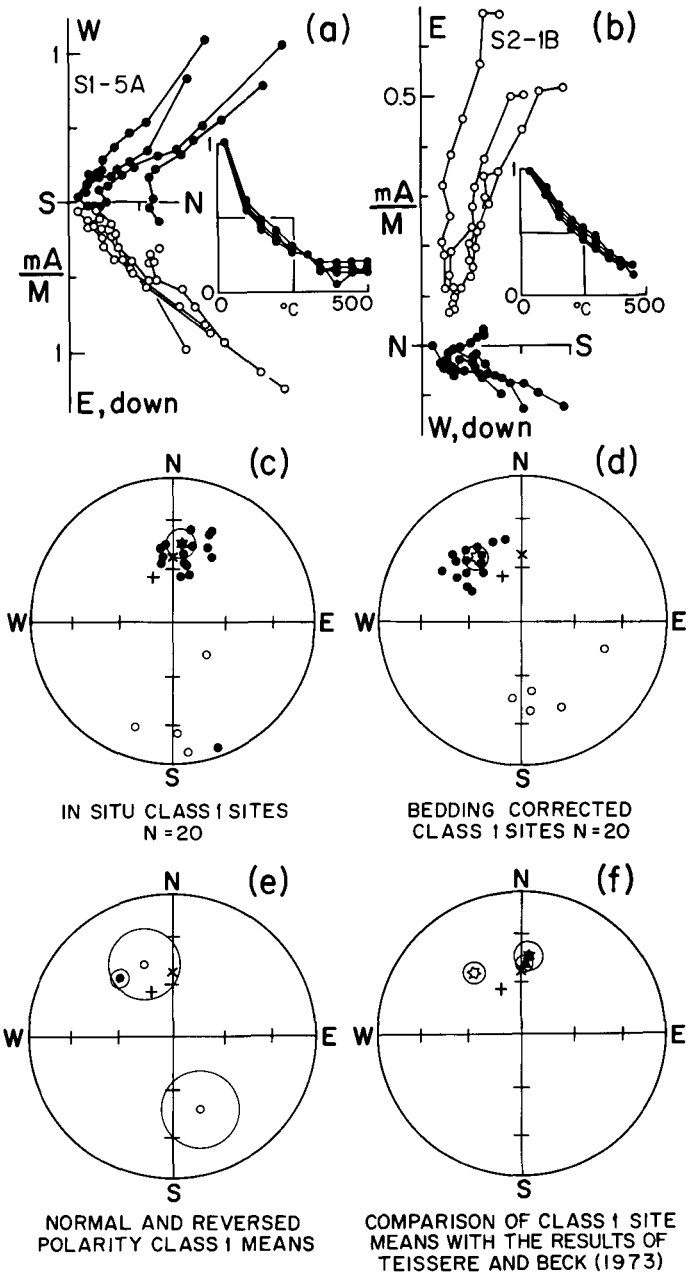


Figure 2. Zijderveld diagrams and stereoplots of some results from this study. a, b: Typical Zijderveld diagrams for two sites, one normal and one reversed in polarity, showing intensity loss and vector change with stepwise heating to 500 °C. Solid circles represent tips of NRM vectors after each heating step plotted on horizontal N, E plane; open circles represent same on vertical (N, down) plane. c, d: In situ and bedding-corrected Class 1 sites of normal (solid circles) and reversed (open circles) polarity. Star = Class 1 mean (in situ mean if filled; bedding-corrected mean if open); + = expected Upper Cretaceous field vector; x = present axial-dipole field vector. e: Normal (solid circle) and reversed (open circle) polarity Class 1 means. Reversed polarity mean also inverted in polarity to show antipodal nature of results. Other symbols as noted above. f: Comparison of in situ and bedding-corrected Class 1 means with axial-dipole and expected Cretaceous field vectors (symbols as noted above) and average field vector (solid circle) calculated by Teissere and Beck (1973) from middle Cretaceous southern California batholith.

Downloaded from geology.gsapubs.org on October 28, 2010

magnetozone (B–) is contained within the Holz Shale. The bottom of the Holz reversed magnetozone is bounded by an intraformational fault that does not have any significant mappable offset (Schoellhamer et al., 1981).

Previous biostratigraphic studies (i.e., Almgren, 1982; Saul, 1982) have tentatively identified Coniacian and Santonian strata in this outcrop area, but stage boundaries have not been precisely located within this Turonian-Campanian sequence. Because the normal polarity interval above the reversed magnetozone is definitely Campanian in age (Fig. 1b), the reversed zone (B–) can be uniquely correlated with the lower Campanian reversed magnetozone Chron 33r (Harland et al., 1982; Fig. 3) and marine magnetic anomaly 33-34. If no significant offset occurred along the fault that forms the lower boundary of the reversed magnetozone, then the lower boundary of this reversed magnetozone may closely approximate the Santonian-Campanian boundary. Strata directly below the reversed interval are those that have been tentatively identified as Coniacian-Santonian in age (Fig. 1b; Almgren, 1982; Saul, 1982). Because no demonstrable large-scale unconformities exist between strata of definite Turonian age (Fig. 1b) and the base of the reversed magnetozone, the paleomagnetic results further strengthen the contention that strata of Coniacian-Santonian age are present in this sequence.

The Campanian reversed magnetozone at Silverado Canyon can also be correlated with an equivalent reversed magnetozone identified by Ward et al. (1983) in Upper Cretaceous strata of the Great Valley Sequence in central California (Fig. 1a). Lower Campanian strata at Silverado Canyon can be correlated with lower Campanian strata in the Great Valley Sequence on the basis of the presence of the biostratigraphically diagnostic ammonite *Submortoniceras chicoense* (Ward et al., 1983; Saul, 1983). This new magnetostratigraphic correlation therefore enhances and independently confirms the soundness of previous lithostratigraphic and biostratigraphic correlations of these units.

TECTONICS

The Class 1 field vectors are the most suitable data with which to calculate an average Late Cretaceous field direction for these strata. The Class 2 field vectors are much more likely to retain vestiges of secondary overprints that may systematically bias any calculated average direction. The average in situ and bedding-corrected field vectors (using only Class 1 results) are I = 44.6°, D = 5.7°, α_{95} = 9.2°, and I = 45.1°, D = –36.4°, α_{95} = 6.7°, respectively (Figs. 2c, 2d; Table 1). The present-day axial-dipole field direction for the Santa Ana Mountains is I = 53.1°, D = 0.0°, and the expected field direction for the Santa Ana Mountains in Late Cretaceous time is I = 62.5°, D = –23.6°, assuming that they were part of the North American stable craton (Irving and Irving, 1982).

The average in situ field vector for the Ladd and Williams Formations is not significantly different from the present-day axial-dipole field direction. This by itself would suggest that the strata do not record a Cretaceous field direction but rather a present-day secondary overprint. However, three lines of evidence indicate strongly that this similarity is coincidental and that the cleaned bedding-corrected paleomagnetic results do retain a good Late Cretaceous remanence. These are (1) that we have recovered a distinctive magnetic polarity sequence that fits previous biostratigraphic age estimates; (2) that the Class 1 normal and reversed polarity field directions are antipodal and therefore pass a reversal test (Fig. 2e); and (3) that the Class 1 field vectors marginally pass a fold test (in situ α_{95} = 9.2°, bedding-corrected α_{95} = 6.7°).

The bedding-corrected average field vector (I = 45.1°, D = –36.4°, α_{95} = 6.7°) is significantly different from the expected Upper Cretaceous North American stable-craton field direction (I = 62.5°, D = –23.6°). The relatively flattened inclination indicates that the Santa Ana Mountains were located at lat 26.6°N in Late Cretaceous time, 17° (1900 km) farther south than they would have been as part of the stable craton (lat 43.8°N). The more westerly declination may indicate that the Santa Ana Mountains have rotated counterclockwise about 10° since Cretaceous time. However, the declination differences are only marginally significant. These results are quite similar to middle Cretaceous paleomagnetic results from the southern California batholith (I = 49.5°, D = 3.0°, α_{95} = 5.0°; Teissere and Beck, 1973), which

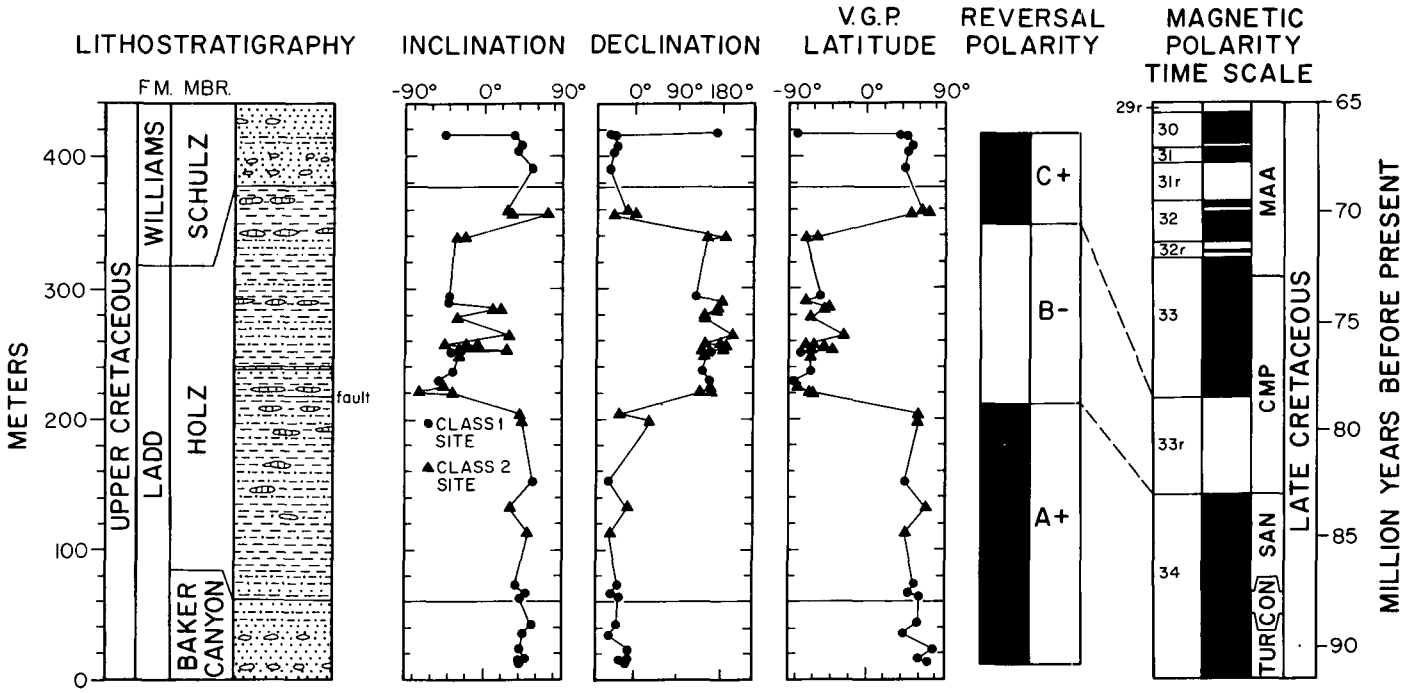


Figure 3. Magnetostratigraphy of Ladd and Williams Formations. Inclination, declination, and calculated virtual geomagnetic pole latitudes are from bedding-corrected Class 1 and Class 2 field vectors. Reversal polarity, determined from these results, contains two intervals of normal polarity separated by large interval of reversed polarity. Single reversed site at top of section may indicate second reversed interval above section, but more data are required to clarify this possibility. Magnetic polarity time scale from Harland et al. (1982).

Downloaded from geology.gsapubs.org on October 28, 2010

TABLE 1. AVERAGED PALEOMAGNETIC RESULTS

	N*	In situ			Bedding corrected		
		I (°)	D (°)	α_{95} (°)	I (°)	D (°)	α_{95} (°)
Class 1 & 2 sites							
All	45	23.1	4.5	9.5	38.3	-19.9	8.2
Class 1 Sites							
Schulz Member	6	49.1	0.1	14.0	43.3	-42.0	16.0
Holz Member	8	33.7	-0.4	19.6	45.4	-35.8	12.8
Baker Canyon Member	6	49.5	21.3	10.0	46.1	-31.6	10.1
Normal polarity	15	49.3	11.1	6.3	44.6	-41.4	5.8
(Norm. pol., Holz only)	(3)	(41.7)	(14.7)	(12.6)	(46.5)	(-43.9)	(10.4)
Reversed polarity	5	-23.5	173.1	31.8	-44.9	159.4	23.0
All	20	44.6	5.7	9.2	45.1	-36.4	6.7

*Number of sites averaged (each site is the average of 4 specimens).

forms the basement of the Santa Ana Mountains. The difference in declination may be attributed to some differential intraterrane rotation as the result of Neogene tectonic translations (Luyendyk et al., 1980; Beck, 1976).

The estimated paleolatitude for the Santa Ana Mountains indicates that the Ladd and Williams Formations were deposited much farther south from the age-equivalent Great Valley Sequence strata in central California (Fig. 1a) than was previously thought (e.g., Bottjer and Link, 1984). This evidence of a significant tectonic displacement of the Peninsular Ranges terrane relative to the North American stable craton has significant implications for the Cenozoic tectonic development of southern California.

CONCLUSIONS

This paleomagnetic study has recovered a unique reversal stratigraphy from Upper Cretaceous strata in southern California. These results allow refinement of the chronology of these strata and better correlation with strata from the Great Valley Sequence in central California. This study also further confirms the usefulness of magnetostratigraphy as a tool for the dating and correlation of sedimentary sequences deposited in active plate margin environments.

Paleolatitude estimates for the Upper Cretaceous strata in southern California clearly document the allochthonous nature of the Peninsular Ranges terrane. The degree of estimated displacement has never been taken into account in tectonic reconstructions of southern California. The lithologic similarity of these strata to the age-equivalent strata of the Great Valley Sequence, even though they were deposited more than 1900 km from each other, typifies the problem of lithofacies correlations in active plate margin environments.

REFERENCES CITED

Almgren, A.A., 1973, Upper Cretaceous foraminifera in southern California, *in* Cretaceous stratigraphy of the Santa Monica Mountains and Simi Hills: Society of Economic Paleontologists and Mineralogists, Pacific Section meeting field trip guidebook, p. 31–44.

——1982, Foraminiferal paleoenvironmental interpretation of the Late Cretaceous Holz Shale, Santa Ana Mountains, California, *in* Bottjer, D.J., Colburn, I.P., and Cooper, J.D., eds., Late Cretaceous depositional environments and paleogeography, Santa Ana Mountains, southern California: Society of Economic Paleontologists and Mineralogists, Pacific Section Field Trip Volume and Guidebook, p. 45–57.

Beck, M.E., Jr., 1976, Discordant paleomagnetic pole positions as evidence of regional shear in the western Cordillera of North America: American Journal of Science, v. 276, p. 694–712.

Bottjer, D.J., and Link, M.H., 1984, A synthesis of Late Cretaceous southern California and northern Baja California paleogeography, *in* Crouch, J.K., and Bachman, S.B., eds., Tectonics and sedimentation along the California margin: Society of Economic Paleontologists and Mineralogists, Pacific Section, v. 38, p. 171–188.

Buck, S.P., 1983, Paleoenvironmental analysis of Late Cretaceous continental slope deposits (Holz Shale Member, Ladd Formation), Black Star Canyon (Santa Ana Mountains), southern California [M.S. thesis]: Los Angeles, California, University of Southern California, 235 p.

Harland, W.B., Cox, A.V., Llewellyn, P.G., Picton, C.A.G., Smith, A.G., and Walters, R., 1982, A geological time scale: Cambridge, Cambridge University Press, 131 p.

Irving, T., and Irving, G.A., 1982, Apparent polar wander paths, Carboniferous through Cenozoic and the assembly of Gondwana: Geophysical Surveys, v. 5, p. 141–188.

Jones, D.L., Howell, D., Coney, P.J., and Monger, J.W.H., 1983, Recognition, character, and analysis of tectonostratigraphic terranes in western North America, *in* Hashimoto, M., and Uyeda, S., eds., Accretion tectonics in the circum-Pacific regions: Tokyo, Terra Scientific Publishing Company, p. 21–35.

Lowrie, W., and Alvarez, W., 1981, One hundred million years of geomagnetic polarity history: Geology, v. 9, p. 392–397.

Luyendyk, B.P., Kamerling, M.J., and Terres, R., 1980, Geometric model for Neogene crustal rotations in southern California: Geological Society of America Bulletin, v. 91, p. 211–217.

Matsumoto, T., 1959, Upper Cretaceous ammonites of California, Part II: Kyushu University Faculty of Science Memoirs, ser. D (Geology), Special Volume I, 172 p.

——1960, Upper Cretaceous ammonites of California, Part III: Kyushu University Faculty of Science Memoirs, ser. D (Geology), Special Volume II, 204 p.

Popenoe, W.P., 1942, Upper Cretaceous formations and faunas of southern California: American Association of Petroleum Geologists Bulletin, v. 26, p. 162–187.

Popoenoe, W.P., Imlay, R.W., and Murphy, M.A., 1960, Correlation of the Cretaceous formations of the Pacific Coast (United States and northwestern Mexico): Geological Society of America Bulletin, v. 71, p. 1491–1540.

Saul, L.R., 1982, Water depth indications from Late Cretaceous mollusks, Santa Ana Mountains, California, *in* Bottjer, D.J., Colburn, I.P., and Cooper, J.D., eds., Late Cretaceous depositional environments and paleogeography, Santa Ana Mountains, southern California: Society of Economic Paleontologists and Mineralogists, Pacific Section Field Trip Volume and Guidebook, p. 69–75.

——1983, *Turritella* zonation across the Cretaceous-Tertiary boundary, California: University of California Publications in Geological Sciences, v. 125, 165 p.

Schoellhamer, J.E., Vedder, J.G., Yerkes, R.F., and Kinney, D.M., 1981, Geology of the northern Santa Ana Mountains, California: U.S. Geological Survey Professional Paper 420-D, 109 p.

Sissingh, W., 1977, Biostratigraphy of Cretaceous calcareous nannoplankton: Geologie en Mijnbouw, v. 56, p. 37–65.

——1978, Microfossil biostratigraphy and stage-stratotypes of the Cretaceous: Geologie en Mijnbouw, v. 57, p. 433–440.

Sundberg, F.A., and Cooper, J.D., 1978, Late Cretaceous depositional environments, northern Santa Ana Mountains, southern California, *in* Howell, D.G., and McDougall, K.A., eds., Mesozoic paleogeography of the western United States: Society of Economic Paleontologists and Mineralogists, Pacific Section, Pacific Coast Paleogeography Symposium 2, p. 535–546.

Teissere, R.F., and Beck, M.E., Jr., 1973, Divergent Cretaceous paleomagnetic pole position for the southern California batholith, U.S.A.: Earth and Planetary Science Letters, v. 18, p. 296–300.

Vedder, J.G., Howell, D.G., and McLean, H., 1983, Stratigraphy, sedimentation, and tectonic accretion of exotic terranes, southern Coast Ranges, California: American Association of Petroleum Geologists Memoir 34, p. 471–496.

Ward, P.D., Verosub, K.L., and Haggart, J.W., 1983, Magnetic anomaly 33-34 identified in the Upper Cretaceous of the Great Valley Sequence of California: Geology, v. 11, p. 90–93.

ACKNOWLEDGMENTS

Supported by National Science Foundation Grant EAR-8213202, Mobil Oil Corporation, and Society of Sigma Xi. We thank L. Kaplan and J. Bannon for assistance in the field and A. A. Almgren and L. R. Saul for critical appraisal of earlier versions of this paper.

Manuscript received February 7, 1985
Revised manuscript received May 24, 1985
Manuscript accepted June 10, 1985